

LAINGSBURG, South Africa

WMO: 687230

Lat: 33.1908S

Lon: 20.8625E

Elev: 2152

StdP: 13.59

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.4	35.6	17.2	14.2	64.5	20.8	16.9	62.4	23.7	56.9	21.5	56.5	1.3	330	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	96.9	63.7	93.6	63.6	90.8	62.9	68.9	83.9	67.5	83.5	66.5	82.6	9.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	98.1	74.5	62.4	91.2	74.7	60.6	85.7	74.2	34.3	84.9	33.2	82.9	32.3	82.4	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	16.9	14.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	77.7	77.7	73.6	65.8	57.9	51.5	50.4	53.7	58.5	66.4	70.2	74.9
	DBStd	11.29	5.86	5.40	6.41	6.51	4.82	4.49	4.94	4.74	5.40	7.12	6.76	5.72
	HDD50	108	0	0	0	0	3	31	55	16	3	0	0	0
	HDD65	1822	2	0	9	70	227	406	451	352	205	70	27	3
	CDD50	5502	858	776	731	473	247	76	69	129	259	507	606	771
	CDD65	1742	394	357	275	93	6	1	0	1	11	112	183	309
	CDH74	15142	3691	3163	2197	699	91	15	16	35	134	983	1502	2616
	CDH80	6338	1719	1438	878	222	13	0	0	5	28	363	567	1105
Wind	WSAvg	6.0	7.1	6.5	6.1	5.3	4.8	5.0	5.0	5.5	6.2	6.9	7.2	7.0
Precipitation	PrecAvg	6.7	0.5	0.6	0.6	0.7	0.7	0.7	0.6	0.5	0.2	0.5	0.6	0.7
	PrecMax	10.2	2.8	2.0	2.3	3.4	2.2	1.7	1.5	1.4	0.7	2.4	2.4	2.9
	PrecMin	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.5	0.7	0.6	0.6	0.9	0.6	0.5	0.4	0.4	0.2	0.6	0.6	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.9	100.1	96.8	90.7	80.9	76.7	76.4	79.7	84.1	95.7	95.5	99.7
		MCWB	64.6	64.2	63.3	61.6	57.0	56.0	54.5	53.2	55.2	62.1	62.2	63.5
	2%	DB	96.3	95.4	92.1	86.0	76.4	69.6	70.6	72.6	77.6	88.6	90.2	93.9
		MCWB	63.9	63.9	64.2	61.1	55.9	51.8	51.4	52.1	54.2	59.3	60.5	63.2
	5%	DB	92.6	91.7	88.5	81.8	72.4	64.7	65.6	68.2	73.4	83.7	86.6	90.1
		MCWB	63.8	63.9	63.1	60.6	54.2	51.1	50.2	51.3	53.5	57.9	60.0	62.8
	10%	DB	89.1	88.5	85.0	77.5	68.7	61.0	61.7	64.3	69.5	79.4	82.8	86.4
		MCWB	63.3	63.7	62.7	59.3	53.7	49.9	48.9	49.5	52.0	57.0	59.3	61.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.0	70.2	70.1	66.4	60.5	57.0	54.8	57.5	61.3	64.6	67.5	68.2
		MCDB	85.8	83.0	84.1	81.1	71.7	68.0	72.5	70.1	70.5	84.5	84.2	85.5
	2%	WB	68.4	68.8	67.9	64.0	58.5	55.0	52.8	55.2	57.0	62.3	64.8	66.7
		MCDB	85.1	83.0	82.6	77.1	70.2	63.5	65.8	67.9	72.4	81.7	79.8	84.0
	5%	WB	66.9	67.6	66.3	62.5	56.7	53.2	51.4	53.2	54.6	60.5	62.7	65.4
		MCDB	84.3	82.9	81.3	74.8	69.1	63.1	62.2	64.4	70.0	78.3	80.1	82.9
	10%	WB	65.6	66.3	64.7	61.0	55.0	51.3	50.0	51.0	52.8	58.8	60.9	64.0
		MCDB	83.5	81.6	80.1	74.7	66.7	60.4	59.9	63.1	67.5	75.9	78.3	81.2
Mean Daily Temperature Range	5% DB	MDBR	21.8	21.0	19.7	19.5	20.1	20.0	21.7	20.7	20.2	21.4	21.4	21.6
		MCDBR	26.2	25.0	24.5	25.8	28.5	27.5	30.3	28.6	27.1	28.5	26.9	26.3
		MCWBR	7.0	6.8	6.7	8.9	12.0	14.0	14.7	13.1	11.6	9.5	7.9	6.7
	5% WB	MCDBR	22.3	20.5	20.3	20.0	22.5	22.9	24.4	23.8	24.2	24.4	22.9	21.8
		MCWBR	7.1	6.3	6.6	7.6	10.4	12.3	12.5	11.6	10.8	8.6	7.8	6.3
Clear-Sky Solar Irradiance	taub	0.327	0.328	0.320	0.298	0.283	0.272	0.275	0.298	0.312	0.331	0.316	0.324	
	taud	2.544	2.544	2.559	2.626	2.649	2.651	2.632	2.536	2.469	2.446	2.539	2.545	
	Ebn at Noon	321	314	305	296	285	281	285	291	303	309	322	323	
	Edh at Noon	35	33	31	26	22	21	22	27	33	36	35	35	
All-Sky Solar Radiation	RadAvg	2622	2353	1919	1429	1060	902	1000	1268	1726	2149	2494	2676	
	RadStd	75	67	72	70	55	56	56	57	83	94	91	120	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page